

Hybrid Biomedical Image-Analysis Pipeline for Nuclei Segmentation

[GitHub](#)

1. Overview

This report describe a hybrid biomedical image-analysis pipeline on a synthetic fluorescence-microscopy nuclei dataset (256×256 RGB, DAPI-style; 80 train / 20 val / 12 test pairs, four density regimes, plus 4 corrupted test variants). Stages: (1) local VLM description (llava via Ollama), (2) classical Otsu+regionprops with text-only LLM interpretation, (3) PyTorch U-Net segmentation, (4) hybrid U-Net→features→structured JSON/narrative. The Central question is: when is a numbers-first, auditable approach more trustworthy than a direct AI description, and where do numeric approaches themselves break down?

2. Task 1: Data Preparation and Multimodal LLM Description

Images were converted to grayscale-derived tensors and already verified at the target resolution of 256x256 (for robustness, the resize logic was not removed). The three images from each of the four density regimes were sampled, and a pixel-intensity histogram was plotted for each.

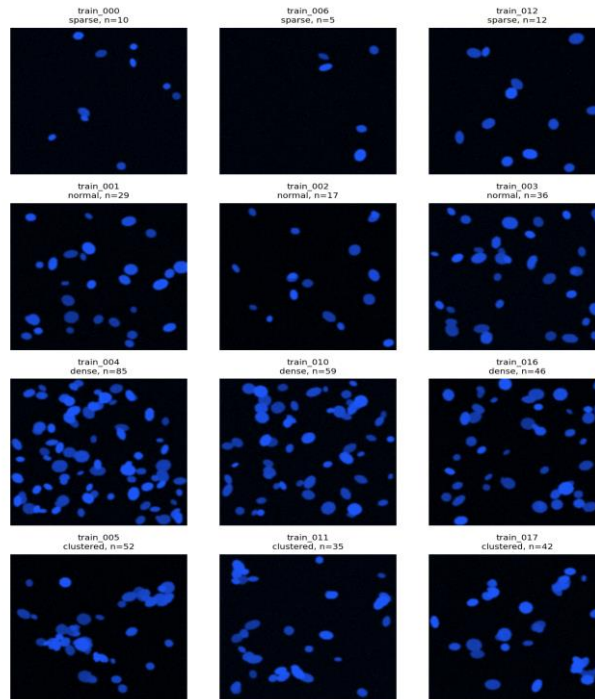


Figure 1: Samples across four density regimes with ground-truth counts (metadata.csv).

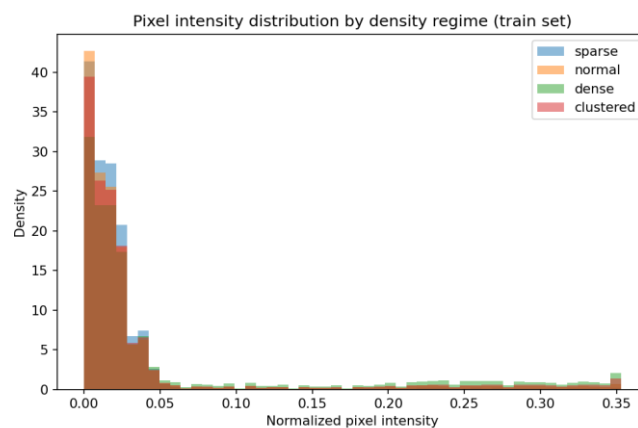


Figure 2: Intensity distribution by density regime background-dominated near 0; regimes differ structurally/spatially, not by intensity.

2.1 Optimized Prompts

Naive prompt:

"Describe this image."

Engineered prompt:

"You are assisting with descriptive image analysis for a research pipeline, not clinical diagnosis. Look at this microscopy image and describe what you observe. Respond ONLY with valid JSON in exactly this format: {"modality": ..., "tissue_type": ..., "notable_features": ..., "image_quality": ...}. If you are not confident about any field, write "uncertain" for that field instead of guessing. Do not include any text outside the JSON object."

2.2 Results

The naive prompt worked absolutely no better and yielded an image labeled as “night sky” and “stars” as a generic pattern match with no biomedical content whatsoever. The engineered prompt correctly identified “fluorescence microscopy” in all 3 runs and the correct labeling of tissue_type “uncertain” as specified, but outputs were non-deterministic (varying wording and formatting in the JSON-fence), and one run hallucinated “green spots” on an objectively blue image a color-grounding failure.

3. Task 2: Classical Features and LLM Interpretation

Otsu thresholding, followed by small object removal, morphological closing and small-hole removal, was applied to the demo image (train_000, sparse regime, ground truth n=10). Connected components were labeled and a regionprops_table computed (area, eccentricity, solidity, mean intensity, perimeter, axis lengths, extent).

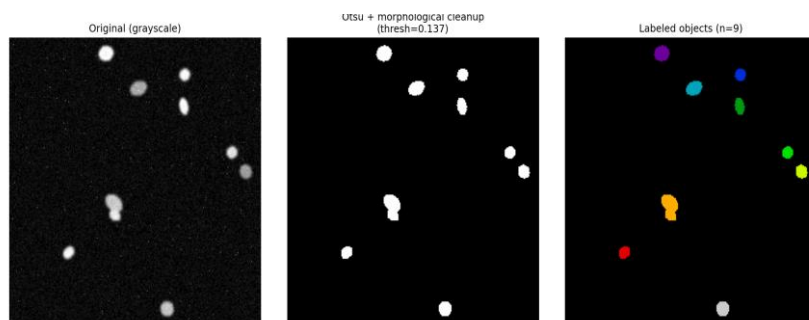


Figure 3: Otsu (thresh=0.137) + cleanup + labeled components, train_000.

Detected 9 objects vs. ground truth 10, the miss is a merged pair of touching nuclei forming one elongated component (area=120), a textbook limitation of intensity-threshold segmentation on touching instances.

3.1 Numbers-First LLM Prompt and Output

The regionprops table was converted to a plain-language numeric summary (counts, area, eccentricity, solidity, intensity, extent) and passed to the LLM as text only, no image shown ,requesting JSON only ({n_objects, density_class, shape_regularity, quality_flag}).

Output: n_objects=9 (correct), density_class="normal", shape_regularity="mostly round and regular" (matches mean eccentricity 0.493), quality_flag="no concerns" , no claims beyond the supplied numbers, unlike Task 1's naive-prompt hallucination.

3.2 Comparison: Task 1 vs Task 2

Aspect	Task 1 (naive VLM)	Task 1 (engineered VLM)	Task 2 (numbers-first)
Domain grounding	Failed (“night sky”)	Correct (microscopy)	Correct (by construction)
Determinism	N/A (not repeated)	No wording/format varies	Deterministic input; LLM interpretation still varies in wording
Verifiability	None	Low (free text)	High: numbers traceable to source pixels
Hallucination risk	Severe	Moderate (colour error observed)	Low: constrained to stated numbers

4. Task 3: U-Net Segmentation

A standard PyTorch U-Net (4 encoder/decoder levels, base width 32, skip connections, BCE+Dice loss, Adam + ReduceLROnPlateau) was trained from scratch for 40 epochs on 80 train pairs (grayscale, batch size 8), evaluated on the 20-image validation split.

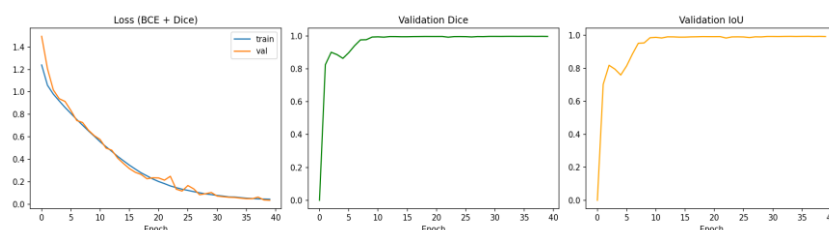


Figure 4: Train/val loss, val Dice, val IoU over 40 epochs.

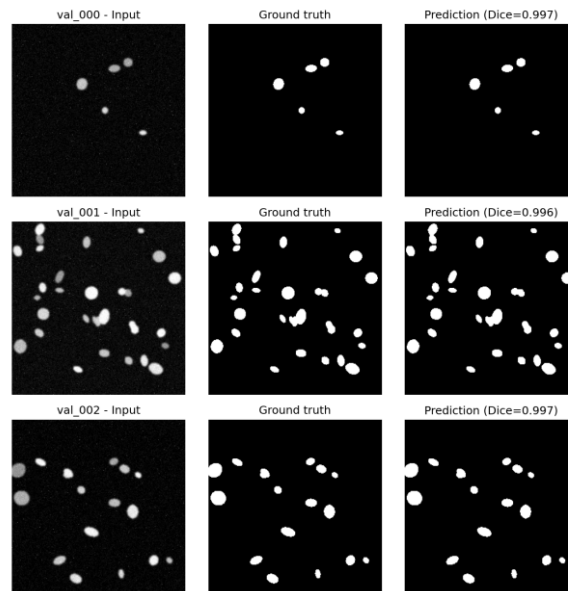


Figure5: Input / ground-truth / prediction for 4 validation images, including a dense case (val_002).

4.1 Evaluation Metrics

Metric	Value
Best validation Dice	0.9972
Validation IoU (at best-Dice epoch)	0.9945
Training epochs	40
Loss function	BCE + Dice (combined)

Loss and Dice converge smoothly, no overfitting (train/val track closely). Predictions are near-pixel-perfect, correctly separating most touching nuclei even in the dense val_002 case.

5. Task 4: Hybrid Pipeline on Test Set

For each of 12 unseen test images, U-Net produced a binary mask, run through the same regionprops pipeline as Task 2. The resulting numerical features were summarized and passed to LLM (text-only). The LLM generated a structured JSON response ({image_id, n_objects, mean_area, density_class, quality_flag}) together with narrative sentence. The pipeline retained the computed numerical results and raw LLM response for each image and the 12 records were aggregated into a DataFrame and exported as task4_hybrid_results.csv.

5.1 Example JSON Record and Narrative (test_000)

{"image_id": "test_000", "n_objects": 8, "mean_area": 190.9, "density_class": "sparse", "quality_flag": "no concerns"} narrative (paraphrased): "A sparse field of well-separated nuclei, no segmentation concerns." (exact wording varies run-to-run, consistent with Task 1's non-determinism.)

5.2 Instance-Count Accuracy vs Ground Truth

image_id	U-Net-derived count	Ground truth	Error	True density
test_000	8	8	0	sparse
test_001	14	19	-5	normal
test_002	26	31	-5	normal
test_003	19	21	-2	normal
test_004	43	75	-32	dense
test_005	17	49	-32	clustered
test_006	8	8	0	sparse
test_007	23	30	-7	normal
test_008	23	28	-5	normal
test_009	21	24	-3	normal
test_010	38	78	-40	dense
test_011	20	30	-10	clustered

Mean absolute count error: 11.75. Count error is near-zero for ground truth sparse images and grows sharply as ground truth density increases, reaching (up to -40) on the densest test image.

5.3 Robustness Extension: Corrupted Images

As an extension, two test images (test_000 sparse, test_004 dense) were corrupted with blur and low-contrast variants and re-run through the trained U-Net without retraining.

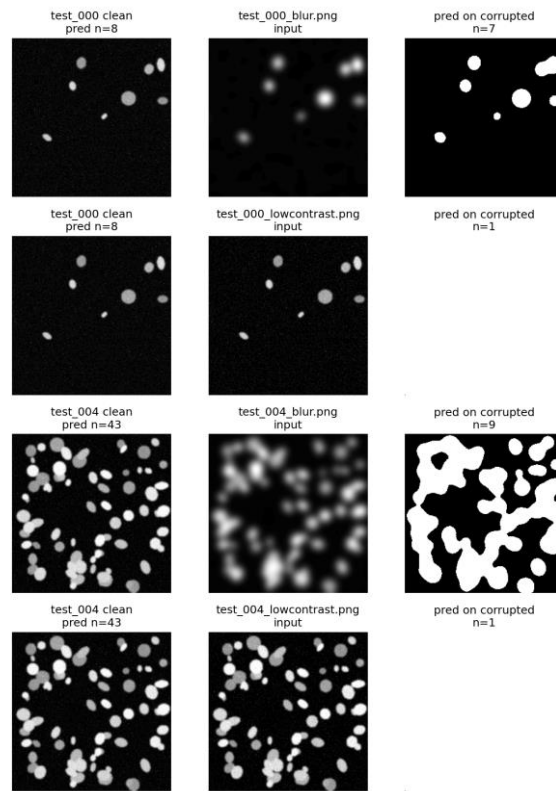


Figure 6: Clean vs. corrupted input and resulting prediction, blur/low-contrast, test_000 and test_004.

Image	Corruption	Ground truth n	Clean pred n	Corrupted pred n	Degradation
test_000	blur	8	8	7	1
test_000	low contrast	8	8	1	7
test_004	blur	75	43	9	34
test_004	low contrast	75	43	1	42

Two failure modes emerged. Blur causes progressive instance fusion: neighbouring nuclei merge into fewer, larger connected components, proportional to local density. Low contrast causes near-total collapse ($n=1$ both cases): the model's learned intensity threshold doesn't generalize to the shifted distribution, collapsing the mask to one large region. Corruption becomes detectable earliest at the mask stage (Stage 3) downstream regionprops/LLM stages inherit the error silently, since regionprops on a collapsed mask still looks internally consistent. An upstream blur/contrast quality gate on the raw input is needed to catch this reliably.

6. Critical Discussion

It is clear by now that the performance of the single-image metrics (VLM plausibility, sparse-case Otsu accuracy, U-Net Dice) is quite good on their own, but all of them suffer from the same problem: they do not explicitly model instance separation, whereas Otsu+connected-components does. The problem of touching nuclei with Otsu is that the intensity is not enough to distinguish between two touching blobs and one blob. At the connected component stage, U-Net faces the same challenge, which is that Dice evaluates the overlap at the level of pixels rather than the number of instances: a near-perfect mask can merge adjacent nuclei into a single connected component. This is explained by excellent Dice (0.997) and systematically undercounted test instances (mean error 11.75) that increases with density. The single most powerful move to take next is to directly solve this issue via watershed post-processing or using an instance-aware architecture (StarDist, Mask R-CNN). The robustness extension reinforces this: corruption happens at the segmentation phase, and any subsequent phase of the pipeline gets its corruption silently.

7. Answers to Assignment Questions

Q1. Which description is more useful/trustworthy direct VLM (Task 1) or numbers-first (Task 2)?

The numbers-first (Task 2) description is more trustworthy: every claim it makes is directly traceable to a regionprops value computed from the image, so an incorrect statement can be checked and explained. The direct VLM description (Task 1) is more immediately readable and can surface qualitative context a numeric summary cannot (e.g. general visual impression), but it is fundamentally unverifiable free text, and the naive-prompt run demonstrated it can fail completely (misreading microscopy as a night sky) with no signal that anything went wrong. For a pipeline aiming at auditability, Task 2's approach is preferable as the source of truth, with the VLM description used only as a supplementary, clearly-labeled qualitative note.

Q2. Did U-Net improve on Otsu? Give an example where each did better.

Yes, substantially, on pixel-level accuracy: U-Net achieved Dice 0.997 vs. Otsu's simpler intensity threshold. On train_000 (sparse), Otsu detected 9/10 true objects, missing one pair of touching nuclei merged into a single connected component U-Net's prediction panels (Figure 5) show comparable or better separation on similarly touching nuclei in denser images such as val_002. However, at the instance-counting level (Task 4), U-Net's advantage narrows: on the densest test images (test_004, test_010), the U-Net-derived object count still undercounts by 32-40 objects because connected-component labeling on its predicted mask has the same instance-fusion issue as Otsu, just at a higher density threshold before it appears. In short: U-Net is unambiguously better at the pixel/mask level; both methods share the same instance-separation weakness at high density.

Q3. Report Dice and IoU. What do they mean and where does the model make mistakes?

Best validation Dice = 0.9972, validation IoU = 0.9945. These measure pixel-level overlap between predicted and ground-truth masks (Dice weights overlap more forgivingly than IoU, but both are near 1, indicating the model reproduces the correct foreground/background pixels almost exactly). The model's mistakes are not visible in these pixel-level numbers; they appear only when the predicted mask is converted to discrete object instances via connected-component labeling. The model tends to merge touching or closely-packed nuclei into single connected regions in dense and clustered images (test_004, test_005, test_010, test_011 in Task 4), producing correct-looking masks with undercounted instances. It is also sensitive to input degradation it never saw during training: low-contrast inputs cause the predicted mask to collapse into a single dominant region (Section 5.3).

Q4. Where can the LLM hallucinate, and what reduces the risk? Why keep JSON as the source of truth?

The LLM can hallucinate at two points: (1) direct-image description (Task 1) e.g. the naive prompt's "night sky" and the engineered prompt's color hallucination ("green spots" on a blue image); (2) numeric interpretation (Tasks 2/4), where it could state a `density_class` inconsistent with the supplied numbers. Mitigations: anchoring as descriptive-not-diagnostic; a fixed JSON schema that makes deviations easy to detect; permitting "uncertain" as an escape hatch; and, most importantly, giving the LLM only numbers (not images) at interpretation, removing visual hallucination entirely. Structured JSON provides a consistent format for LLM's Interpretation, while the computed numerical features remain auditable source of truth because they are directly generated from segmentation mask and regionprops measurement.

Q5. Would you trust any part of this system clinically? What single change would most improve trustworthiness?

No part of this system should be trusted for real clinical decisions yet. The dataset is fully synthetic with clean intensity separation unlike real tissue; U-Net trained on only 80 images from one generator; and Sections 5.2–5.3 show silent failure (undercounting, mask collapse) in exactly the high-density, low-quality conditions real specimens are likely to produce, with no built-in signal that this happened. The most trustworthy element is the regionprops table itself (deterministic, auditable, traceable to a mask) but only as good as the mask it came from. The single change with most impact: instance-aware segmentation (watershed or StarDist/Mask R-CNN) combined with an automated image-quality gate that flags low-quality inputs before they silently produce a plausible-but-wrong JSON record.

8. Limitations and Future Work

Limitations: (1) synthetic dataset may not capture real-microscopy noise/staining diversity; (2) Dice/IoU don't capture instance-separation quality, the metric that matters for counting; (3) no automated failure detection short of manual inspection; (4) small sample (80 training images). Future work: instance-aware segmentation, blur/contrast training augmentation, and an automated image-quality gate.

References

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